

Virtual Try-On Models

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I built Virtual Try-On on my own laptop, an RTX 4070 with eight gigs. I wanted to say why each option is there, which clothes it handles better, and what this card can actually do. No retraining — just running the models. Seven weights load here. Gemini and the Hugging Face Space are for when the GPU is busy or too small for FLUX: the same person and garment photos go out, and the heavy weights never sit on the laptop.

9 menu options · 7 local weights + 2 cloud	8 GB SD fits on this card	Cloud Gemini with a key · CatVTON on HF Space	5.425 best paper FID on VITON- HD
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How I put it together

What a checkpoint is

The weight file — the numbers the network learned. The paper says weights; Hugging Face says checkpoint. There are nine options in the menu, but not nine separate networks. Mix, VITON-HD, and DressCode share one UNet and only swap the attention file — about 198 MB. Mask-Free is different: an 8-channel Pix2Pix UNet. The three FLUX options bring Fill or LoRA weights. Gemini and the Space load nothing locally.

Input is always simple: a photo of a person, a photo of a garment, and whether I want the top, the bottom, or the full look. I center-crop the person onto a 3:4 canvas and letterbox the garment with a white border to the same size. The paper uses DensePose and SCHP for the mask; I used SegFormer so it would start on Windows without Detectron2. Then one of four paths runs: SD inpaint, Pix2Pix, FLUX, or the cloud. If something breaks, it no longer stays in the terminal — a red bar at the top shows the same text. Generate locks while a job is running, and Stop only lights up when something is actually in flight.

SD 1.5 inpaint — three attention weights One shared UNet, about eight gigs. Switching models only moves that 198 MB of attention. I set guidance to 2.5. No Hugging Face login. Mix · VITON-HD · DressCode	Mask-Free — Pix2Pix Eight-channel UNet, not inpaint. Person and garment sit side by side. Guidance is still 2.5. No mask goes into the network; if the request is Top, I restore trousers and the face from the original photo afterward. catvton-maskfree · 8 channels	FLUX and cloud FLUX.1-Fill-dev is gated and its guidance is 30. On a small card it goes Offload or 4-bit, but slowly. Gemini with an AI Studio key and the public CatVTON Space on ZeroGPU load no weights on the laptop. FLUX · Alpha · Beta · Gemini · HF Space
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Quality comes from the menu. Tiny is 256×384 and 16 steps, Fast 384×512 and 20 steps, Balanced — what I use day to day — 576×768 and 30 steps. Quality is 768×1024 and 50 steps, Studio 1024×1280 and 36 steps. If VRAM is Auto and the card is under 24 GB, FLUX switches itself to Offload or 4-bit.

What these four numbers mean

The paper reports four scores. The first two compare a pair of photos: my output next to a real photo of the same person in the same clothes. The other two compare the whole set of outputs to the whole set of real photos, even when there is no exact pair.

Metric	What it says	Better	Range	Where it lies
SSIM	Window by window, how close it is to the real photo	Higher, near 1	0 to 1	A blurry but aligned image can beat a sharp one that is slightly shifted
LPIPS	How much difference a network “eye” feels	Lower	Usually 0.04 to 0.20	Depends on that network; it does not score the face on its own
FID	How close the whole set of outputs is to real photos	Lower	Unitless; here about 3 to 15	The number jumps if the eval code or the sample count changes
KID	Same idea, calmer on a small set	Lower	Unitless	Still lives in Inception space

SSIM

It compares window by window with the real photo: brightness, contrast, edges. The score sits between zero and one. For me a high SSIM means the face, hair, and background stayed put — not that the garment looks nicer. In the paper table, CatVTON inpaint on VITON-HD reaches 0.8704. Close, not identical.

LPIPS

Not raw pixels. Patches go through a network like VGG and the distance is measured there. The eye cares more about texture and folds than an RGB gap; this approximates that. Lower is better. If the garment sits right but the color is a bit off, LPIPS complains more than SSIM. Inpaint on VITON-HD is 0.0565.

FID

This one is not for a single pair. All the real photos and all the outputs go through Inception, each pile gets a mean, and you see how far the two clouds sit. Low FID means the outputs look like real photos. The catch is that changing the eval code or the number of images moves the number. That is why I do not mix the community FLUX FIDs into the paper table.

KID

Same space, without FID’s statistical assumption. On a few thousand images it usually sits still. The paper reports both because everyone knows FID and KID jumps less on these sets. On paired VITON-HD, CatVTON hits 0.411 and OOTDiffusion 4.086.

What I do before the model

All nine start from the same person, garment, and garment type. From here their paths split.

Step	What it does	Where in the code
1. Canvas	I center-crop the person; I pad the garment with a white border to the same size	resize_and_crop / padding
2. Body parse	SegFormer splits clothes into 18 parts: face, upper clothes, trousers, hands. It stays on CPU so the GPU stays free	ClothMasker
3. Mask	For Mix and the other inpaints, the region grows with Top / Bottom / Full look and the face and hair stay. Mask-Free does not send this to the UNet	build_agnostic_mask
4. Edge	I blur the inpaint mask a little so the garment edge does not tear	blur_mask
5. Generate	SD inpaint, Pix2Pix, FLUX, Gemini, or the Hugging Face Space	catvton / flux_tryon / cloud
6. Lock	On Mask-Free Top I restore trousers, legs, shoes, and the face from the original photo	composite_tryon

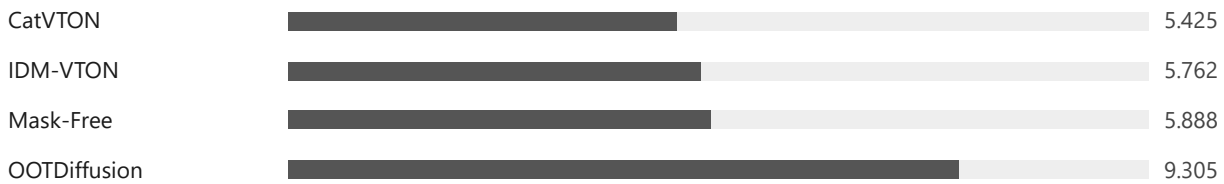
The paper has DensePose and SCHP. I used SegFormer because Windows install was simpler. Mask-Free is for when the mask catches on belts and busy prints — with the real Pix2Pix UNet, not the 9-channel inpaint one.

Paper numbers

Just keep this

FID and KID: lower is better. SSIM: higher is better. LPIPS: lower is better. Paired means there is a reference photo. Unpaired only says the outputs look like real photos. I kept the community FLUX numbers apart; they were not taken with the paper's method.

Paired FID on VITON-HD — lower is better



Paper Table 1, paired VITON-HD. FID has no unit; lower is better.

Paper time at 1024×768

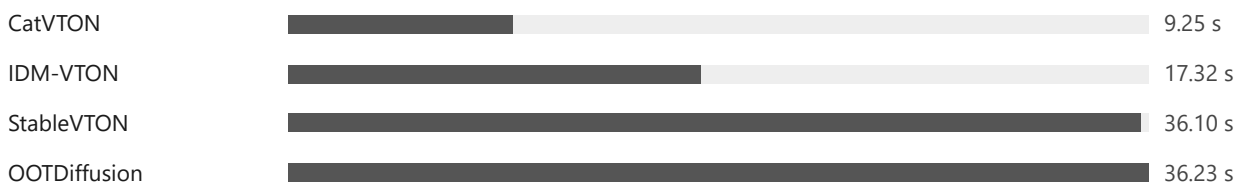
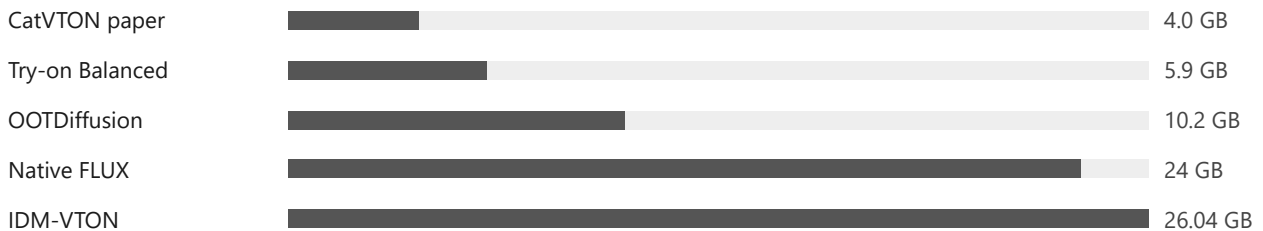


Table 2 of the same paper. Taken on a datacenter card, not my laptop.

Memory — gigabytes



Paper Table 3, plus Balanced on my own laptop.

Paper table on VITON-HD

Method	SSIM ↑	Paired FID ↓	Paired KID ↓	LPIPS ↓	Unpaired FID ↓	Unpaired KID ↓
CatVTON inpaint	0.8704	5.425	0.411	0.0565	9.015	1.091
CatVTON Mask-Free	0.8701	5.888	0.513	0.0613	9.287	1.168
IDM-VTON	0.8499	5.762	0.732	0.0603	9.842	1.123
OOTDiffusion	0.8187	9.305	4.086	0.0876	12.408	4.689
GP-VTON	0.8701	8.726	3.944	0.0585	11.844	4.310
StableVTON	0.8543	6.439	0.942	0.0905	11.054	3.914
LaDI-VTON	0.8603	11.386	7.248	0.0733	14.648	8.754

Paper table on DressCode

Method	SSIM ↑	Paired FID ↓	Paired KID ↓	LPIPS ↓	Unpaired FID ↓	Unpaired KID ↓
CatVTON inpaint	0.8922	3.992	0.818	0.0455	6.137	1.403
CatVTON Mask-Free	0.9016	4.779	1.297	0.0452	7.400	2.619
OOTDiffusion	0.8854	4.610	0.955	0.0533	12.567	6.627
IDM-VTON	0.8797	6.821	2.924	0.0563	9.546	4.320
GP-VTON	0.7711	9.927	4.610	0.1801	12.791	6.627
LaDI-VTON	0.7656	9.555	4.683	0.2366	10.676	5.787

The paper’s DressCode weight only learned on DressCode. Mix, which I set as the default, learned on about 48 thousand mixed pairs at 1024 pixels and has no separate table in the paper.

Why it is light

Method	Total params (M)	What it learned (M)	Memory (GB)	Time 512×384	Time 1024×768
CatVTON	899	49.57	4.00	2.58 s	9.25 s
StableVTON	1608	501	7.87	12.17 s	36.10 s
OOTDiffusion	2191	1719	10.20	4.76 s	36.23 s
IDM-VTON	6238	2871	26.04	12.96 s	17.32 s

Tables 2 and 3. CatVTON dropped the text encoder and ReferenceNet and only trained self-attention.

Community FLUX numbers — a separate story

I do not add these to the table above

Their author reports FID with scale 30 and 30 steps on VITON-HD. Mask, image size, and eval code are not stated as matching the paper.

Weight	Type	Trained on	Reported FID	Test
FLUX Alpha full	Transformer fine-tune	VITON-HD only	5.593	Scale 30 · 30 steps
FLUX LoRA Alpha	LoRA on Fill	VITON-HD only	6.068	Scale 30 · 30 steps
Authors' FLUX	Author LoRA · 37 MB	Not in the paper	Unpublished	Guidance 30 in Virtual Try-On
FLUX Beta	Full fine-tune ~23.8 GB	More garment variety	Unpublished	Guidance 30 in Virtual Try-On

Time on my laptop

Fast 384×512

11 s

Balanced 576×768

19 s

With Mix on an RTX 4070 Laptop 8 GB, September 2026. VRAM about 5.9 of 8.0 GB. I have not timed Quality, Studio, or native FLUX on this card.

Why I put these in

Mix — what I run every day

Day to day I go to Mix. It is the same SD inpaint; the difference is that its attention learned on about 48 thousand mixed pairs at 1024 pixels, tops and trousers both. On my laptop it stays stable at Balanced and needs no login. Print and texture still trail FLUX.

id catvton-mix · zhengchong/CatVTON · mix-48k-1024 · Guidance 2.5

VITON-HD — shirts and jackets

Same engine as Mix; only the attention file changes. It learned on shirts and jackets. I keep it for upper-body catalogs. I do not reach for it if I want trousers or a full outfit.

id catvton-vitonhd · vitonhd-16k-512

DressCode — dresses and variety

I pick this when I want an evening dress or a full set. That paired FID of 3.992 in the paper belongs to this family on DressCode, not to Mix.

id catvton-dresscode · dresscode-16k-512

Mask-Free — busy photos

This one does not give the network a mask. Person and garment sit side by side and the model guesses where the clothes should change. At first I loaded the inpaint UNet by mistake and it stripped the clothes instead of putting them on. Now it is the right Pix2Pix. The next problem was that with Top it sometimes changed the trousers too, so after generation I restore trousers, legs, shoes, and the face from the original photo. On a clean photo it is not better than Mix.

id catvton-maskfree · zhengchong/CatVTON-MaskFree · timbrooks/instruct-pix2pix

Gemini — Google cloud

No local weights. Person and garment go out and a try-on photo comes back. I put the key from aistudio.google.com/apikey in the UI field; it stays on this machine in `.secrets.json` and never goes to git. I try `gemini-3.1-flash-image` first, then `gemini-2.5-flash-image` if that model is missing. I pick this when the GPU is busy or too small for FLUX. Without a key, the error banner shows the same AI Studio link.

id gemini-flash · Google AI Studio key · no local VRAM

Hugging Face — public CatVTON Space

Same cloud idea, but the engine is CatVTON, not Gemini. The authors' Space [zhengchong/CatVTON](https://huggingface.co/zhengchong/CatVTON) is often `RUNTIME_ERROR`. I check what is live and go to [tryitnvirtual/virtual-tryon](https://tryitnvirtual.com/virtual-tryon) or [Nandha2017/virtual-tryon](https://nandha2017.com/virtual-tryon) — both on ZeroGPU A10G. Without a token the quota is thin; one run asked for 180 seconds and only 176 were left. An `hf_` token from huggingface.co/settings/tokens raises the quota and also unlocks gated FLUX downloads.

id hf-catvton · ZeroGPU · optional token

Authors' FLUX — light and folds

The CatVTON authors' own LoRA on FluxFill, about 37 MB. I set guidance to 30. Light and folds beat SD; on eight gigs it only moves with Offload or 4-bit, and it is slow.

id catvton-flux · Guidance 30 · concat person_first

FLUX Alpha — logos and texture

A community LoRA on VITON-HD. I pick this when logos and text on the fabric matter. They report an FID of 6.068; I do not mix it into the paper table.

id catvton-flux-alpha · xiaozaa/catvton-flux-lora-alpha

FLUX Beta — garment variety

It swaps the whole transformer, about 23.8 GB. More garment variety than Alpha, and the heaviest download.

id catvton-flux-beta · xiaozaa/catvton-flux-beta

A quick look

Model	Best use	VRAM	Login	Mask	Strength	Weakness
Mix	Everyday	8 GB	Not needed	Clean	Balanced coverage	Less detail than FLUX
VITON-HD	Upper body	8 GB	Not needed	Clean	Shirts and jackets	Weak on full outfits
DressCode	Dresses and sets	8 GB	Not needed	Clean	Category variety	Trained at 512 px
Mask-Free	Busy photos	8 GB	Not needed	Pix2Pix + trousers lock on Top	Broken masks	Without the lock, it changes the lower body too
Gemini	GPU busy or small	Cloud	AI Studio key	No local mask	Zero VRAM	API cost · photos leave for Google
HF Space	CatVTON in the cloud	Cloud	Optional token	Space mask	No local load	ZeroGPU quota · authors' Space is often down
FLUX	High quality	24 GB+	HF + license	Mask + Fill	Light and fabric	No paper FID
FLUX Alpha	Logos and texture	24 GB+	HF + license	Mask + Fill	Small prints	Upper body only
FLUX Beta	Garment variety	24 GB+	HF + license	Mask + Fill	Wide coverage	Heaviest weight

The hard parts

Windows and weight downloads

Hugging Face’s XET layer kept stalling on Windows. I turned XET off and put a resume-capable curl script in. The SD 1.5 UNet itself is gated too; I try a public mirror first.

Eight gigs versus FLUX

My card is 8 GB. SD 1.5 fits at Balanced. Native FLUX wants 24 GB. Offload and 4-bit make it run, but very slowly. That is why I also put Gemini and the Hugging Face Space in, so the heavy weights never sit on this laptop.

Mask-Free without a trousers lock

Pix2Pix has no inpaint mask. On Top, it once changed the subject's trousers too. After generation I restore trousers, legs, shoes, and the face from the original photo. If the parser misses the trousers, the lock is incomplete.

ZeroGPU quota and the Gemini key

The Space is live and the queue and upload work, but without an hf_ token the guest quota is thin. Gemini does not run at all without a key; the error text has to show at the top, not only in the terminal. I keep the key and token in .secrets.json.

A simpler mask than the paper

The paper has DensePose and SCHP. I put SegFormer on the CPU so the GPU stays free for generation. Install got easier; the mask edge can be weaker.

Non-commercial licenses

CatVTON and SD weights are CC BY-NC-SA 4.0. FLUX.1 [dev] is also non-commercial and gated. Gemini follows Google's Gemini API terms and takes the photos off the machine. If I ever want commercial work, I need a different model or a separate license.

Where they get stuck

Family	Main snag	For me that means
All three SD inpaints	The mask has to be clean	A hard pose or a busy background tears the garment
Mix	No separate table in the paper	I judge quality by eye, not a dedicated FID
VITON-HD / DressCode	Each one is one garment category	Pick the wrong category and quality drops
Mask-Free	Paired FID a bit worse · lower-body leak	Only worth it when the mask is broken; I lock Top
Gemini	Key, and photos go to Google	No key, no generate; cost is on the AI Studio account
HF Space	Authors' Space down · ZeroGPU quota	Without a token the queue may fill or 180 seconds may not be enough
All three FLUX	Gated download and large files	First load can break without a login
FLUX Alpha	Trained on upper body only	Lower body and dresses weaker than Beta
FLUX Beta	Full transformer weight	Switching models empties VRAM and fills it again

When I reach for which

Scenario	First pick	Second pick	Why
8 GB laptop, daily tests	Mix + Balanced	Mix + Fast	Stable, fast, no login
Shirt and jacket catalog	VITON-HD	FLUX Alpha on HF	Upper-body data
Evening dress or full set	DressCode	FLUX Beta on HF	Category variety
Selfie or a torn mask	Mask-Free	Mix with the right garment type	Pix2Pix + trousers lock on Top
GPU busy, or FLUX too slow on 8 GB	Gemini	HF Space	Zero local VRAM; key or token
CatVTON in the cloud, no Gemini	HF Space + token	Local Mix	Authors' Space is often <code>RUNTIME_ERROR</code>
Logos and text on fabric	FLUX Alpha	Authors' FLUX	Graphic detail
Best quality I can get	FLUX on 24 GB	Gemini or Quality with Mix	Higher perceptual ceiling · cloud when VRAM is short
Commercial production	None of the CatVTON/FLUX weights	A licensed model	CC BY-NC-SA and FLUX.1 [dev] are non-commercial; Gemini is separate

Last word

If I only look at the paper numbers, CatVTON inpaint is both light and accurate: paired FID 5.425 on VITON-HD and 3.992 on DressCode, about 4 GB and 9 seconds at 1024 on their card. Mask-Free sits a little behind, but it does not depend on a clean mask.

I put that family on the laptop. With Mix, Fast is about 11 seconds and Balanced about 19. FLUX raises the ceiling, but it also brings login, size, and VRAM. Gemini and the Space are for when the GPU is busy or FLUX is not worth it on this card. Day to day, Mix is enough. I pick Mask-Free when the mask is torn. I pick FLUX or Gemini when fabric detail or a VRAM shortage actually pays for it.

Sources

Table numbers come from Tables 1–3 of the CatVTON paper unless I wrote otherwise. Laptop times are my own with Mix on an RTX 4070, eight gigs, in September 2026. The paper numbers are on an A800 and do not add to mine.

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